

Self-Attention and QKV

Deep Learning

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Contents of This Video

In this video, we will cover:

- Self-attention vs. cross-attention
- The database analogy: Query, Key, Value
- Computing self-attention with QKV
- The scaled dot-product formula
- Why self-attention is powerful

Self-Attention: A Sequence Talks to Itself

Cross-attention: Decoder attends to encoder

Self-attention: Sequence attends to itself

Example: “The cat sat on the mat because it was tired”

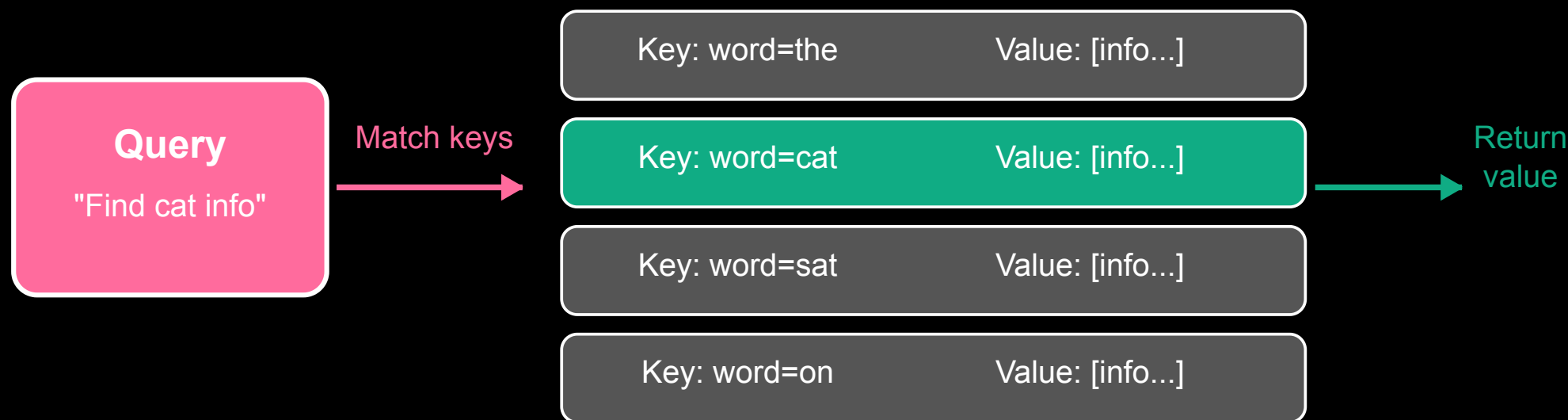
What does “it” refer to?

- Self-attention: “it” looks at all other words
- High attention to “cat” → “it” means “cat”

The Database Analogy

Query, Key, Value = Database lookup

QKV: Query Matches Keys to Retrieve Values
Database



QKV Projections

Each input x_i creates three vectors:

$$q_i = W_Q x_i \quad (\text{Query: what I'm looking for})$$

$$k_i = W_K x_i \quad (\text{Key: what I contain})$$

$$v_i = W_V x_i \quad (\text{Value: what I provide})$$

Three learnable weight matrices: W_Q, W_K, W_V

Scaled Dot-Product Attention

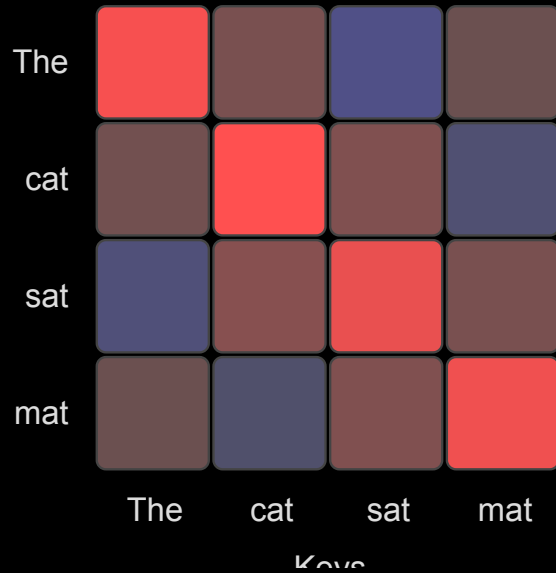
$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^{\top}}{\sqrt{d_k}} \right) V$$

Step by step:

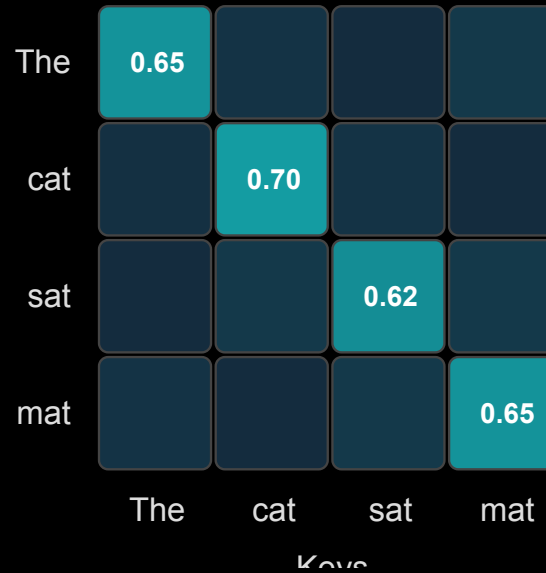
1. **Score:** $QK^{\top}$ gives similarity between each query-key pair
2. **Scale:** Divide by $\sqrt{d_k}$ (key dimension) for stable gradients
3. **Normalize:** Softmax over keys for each query
4. **Combine:** Multiply by V to get weighted values

Self-Attention Example

QK^T (scores)



Softmax (attention)



Result

Each word attends to all words

"cat" query matches "cat" key (high) + others (low)

Output = weighted sum of values

Why Self-Attention is Powerful

Global context: Every position sees every other position

Parallelizable: All attention computed simultaneously (unlike RNNs)

No distance penalty: “cat” can attend to “it” equally well regardless of distance

Learns relationships: Attention patterns capture syntactic/semantic structure

What We've Covered

In this video, we've learned:

- Self-attention: sequence attends to itself
- Query (what I want), Key (what I am), Value (what I give)
- QKV are linear projections of the input
- Scaled dot-product: $\text{softmax}(QK^\top / \sqrt{d_k})V$
- Every position can attend to every other position
- Parallelizable and captures long-range dependencies